Slow Arc — User Guide

For anyone running the installation day to day. No technical knowledge needed.

The controller is a small box that runs **six Slow Arcs**. Each arc has a pair of bulbs that fade up and down, and a motor that turns it slowly. The box remembers everything you set, and keeps running on its own whether or not there is WiFi.

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0. Overview

This control box powers and controls six Slow Arcs at once. It has two modes: Gallery and Performance. In Gallery mode, each artwork runs continuously at its own settings. In Performance mode, all the artworks stay dark until a timed sequence, in which they turn on and off, is started by pressing Play or the sequence button.

The box has both physical controls and a web page that it hosts itself. Every physical control on the box is repeated on the web page. The page shows detailed information about the state of the artworks and the control box itself, and also lets you create, select, edit or delete the stored performance sequences.

Optionally, a music track can be loaded into the web page on a laptop and played in sync with the performance sequence (§9).

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1. The box

> **[photo placeholder]** — `docs/img/front-panel.jpg` The front panel: the mode switch, the sequence button, and the status light.
>
> **[photo placeholder]** — `docs/img/box-installed.jpg` The box in place, showing where the arcs connect.

There are only three things on the outside:

	What it does
Mode switch	Chooses Gallery or Performance (see below). The web page can do the same; whichever was used most recently is in charge.
Sequence button	In Performance mode, starts and stops the choreography. Does nothing in Gallery mode. The web page's Play / Stop button does the same.
Status light	A small coloured light telling you the box is healthy. See §5.

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2. Turning it on

Switch on the power. Then:

1. The status light comes on **solid amber** for a few seconds.
2. The bulbs fade up gently — never all at once, and never instantly. A slow, staggered rise is normal and deliberate; it protects the bulbs and the power supply.
3. The motors start turning.
4. The light settles into a **slow green pulse**. The box is running normally.

That is a box in **Gallery** mode. A box that was left in **Performance** keeps every arc dark and still instead, and the light settles into **blue blinking**: it is waiting for the piece to be started (§3).

Everything comes back exactly as you left it. Brightness, speed, mode, and your sequences are all remembered through a power cut, including a mode chosen on the web page, whatever the switch says. A sequence that was playing does **not** restart by itself: the box comes back in Performance, waiting with every arc dark and still, until someone presses Play or the sequence button.

One exception: if you switched any arcs off from the web page, they come back **on** after a power cycle. That is intentional — a box restarting overnight shouldn't come back with an arc dark that nobody notices.

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3. The two modes

Set with the physical switch on the box **or** the Gallery / Performance buttons at the top of the web page's Control tab. Whichever was used last is in charge, so a box mounted out of reach can be run entirely from the page (§6).

Gallery — each arc simply holds its own settings. Arc 1 stays at its brightness and speed, arc 2 at its own, and so on, indefinitely. This is the normal day-to-day mode.

Performance — the box plays a **sequence**: a timed piece where arcs fade in and out in a set order. The status light turns blue and blinks the moment the mode changes, which is your confirmation it has taken. Press the sequence button (or **Play** on the page) to start the piece; press again (or **Stop**) to stop. Play on the page also switches to Performance if the box was in Gallery.

Staging. Going into Performance takes every arc dark and stops the motors, and they stay that way until the piece is started — so the performance opens from darkness. The page shows the sequence as **staged** while it waits. To stage again after a rehearsal, switch to Gallery and back (the arcs light briefly in between). Pressing Play while the box is in Gallery skips staging and starts the piece at once.

When you stop a sequence, the arcs **hold wherever they are** rather than jumping back. Switch to Gallery to return them to their normal settings.

When the page overrides the switch. If the page sets a mode different from where the switch sits, the page wins until the switch is next flipped. The switch then no longer shows the box's real mode, so the status light adds a **brief white blip every two seconds** on top of its normal pattern, and the page

outlines the front-panel box in red with **overridden by this page** across it. Either flip the switch (twice, if it has to end up where it already is) or choose the mode on the page that matches the switch, and the blip stops.

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4. Getting to the control page

The box hosts its own web page. You reach it from a phone, tablet, or laptop.

If the box is on the gallery WiFi — connect your device to the same network and go to `http://slow-arc.local`. That name follows the box around, so you never need to know or write down the number the router gave it. It works on Macs, iPhones, iPads, and current Windows; on anything that can't resolve it, use the numeric address shown on the box's status line instead, e.g. `http://192.168.1.47`.

The status line prints both, and the `slow-arc.local` line is a link — so once you're on the page from any route, that's where to find the address again.

If the box is not on any network (a new box, or the WiFi is down) — it makes its own network:

- **Network name:** `SlowArc-Setup`
- **Password:** `slowarc123`

Join that from your phone and the control page should open by itself. If it doesn't, go to `http://192.168.4.1`.

This always works, even with no internet in the building, so you can always reach the box on site.

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5. The status light

Light	Meaning
Solid amber	Starting up. Normal for the first few seconds.
Slow green pulse	Healthy, in Gallery mode.
Blue blinking	Healthy, in Performance mode, waiting (arcs dark until the piece starts). Press the sequence button.
Slow blue pulse	Healthy, sequence playing.
Amber pulsing	Looking for WiFi. Harmless — the arcs run regardless.
Fast blue flicker	A software update is being installed. Do not cut the power.
Red flashes	A fault. **Count the flashes** — see below.
White blip every 2 s, on top of any of the above	The web page has overridden the mode switch — the switch isn't showing the real mode. See §3.

The light turns blue the instant the box goes into Performance, so you can confirm the change has taken without waiting to see what the arcs do. It blinks on and off while waiting and settles into a smooth breathing pulse once a sequence is playing. (The blue flicker of a software update never goes fully dark

— that is how you tell the two apart, and updates only happen when someone is deliberately installing one.)

Red flashes repeat in groups with a pause between:

- **2 flashes** — a bulb problem. One or more arcs may not light.
- **3 flashes** — a motor problem. One or more arcs may not turn.
- **4 flashes** — a power problem. Usually means the supply dipped when the bulbs came on.

Count the flashes and pass the number to whoever maintains the box — it tells them where to look. A single red pattern that clears itself after a minute or so following a restart is worth mentioning but not urgent.

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6. The Control tab

The top strip is the box's status at a glance: healthy or faulty, which mode, which sequence is selected and whether it is running, stopped or **staged** (arcs dark, waiting for Play), which WiFi network it is on, and how long it has been on. If the box is still serving its own **SlowArc-Setup** network, an amber line asks you to connect it to a local WiFi network, with a link to **WiFi setup**.

Under the tabs, the **Override physical controls** card has two halves. On the left, **This page** does the job of the front panel: **Gallery** and **Performance** choose the mode, and **▶ Play / ■ Stop** starts and stops the selected sequence. Stopping a running sequence — with **Stop**, or by choosing **Gallery** — asks **Stop the performance?** first, so a stray tap can't end a show. The highlighted button is the box's current mode. On the right, **Front panel** shows the controls on the box itself: which way the mode switch is set, and when the sequence button was last pushed. If the page has overridden the switch, that half is outlined in red and marked **overridden by this page** (§3).

If the strip turns red and says **OFFLINE**, the page has lost contact with the box — it has been powered down, or your phone has left the network. The sliders below grey out and stop responding, because what they are showing is the last reading received rather than anything current. The strip counts up how long it has been out of contact, and comes back by itself within a few seconds of the box returning.

Below the **Performance** box (§7, §9) is the **Set-points** card, with one column per arc. **Note that these values should not be adjusted except on explicit instructions from Conrad or his team.**

- The **blue slider (M)** is motor speed, 0–100.
- The **yellow slider (B)** is bulb brightness, 0–100.
- The **box under the two sliders** is the motor's speed in Hz — the actual step rate, 60 to 8000. It is the real setting; the M slider is just a quick way to move it. Type a number and press Enter (or tap away) to set a speed exactly, e.g. to give two arcs the same rate or to return to a number written down at commissioning.
- The **on / off button** switches that arc off completely — dark and stopped — without losing its settings. Useful if a bulb blows or an arc needs attention mid-show. Press again to bring it back exactly as it was.

Changes take effect as you drag or as you leave the Hz box, and save themselves a second or two later. There is no save button and nothing to confirm.

Set 0 to turn that half off: brightness 0 is dark, speed 0 is stopped. A typed speed between 1 and 59 Hz is raised to 60 — below that the motor is switched off instead, so there is no crawling-but-not-quite-stopped state.



7. Watching a sequence play

In Performance mode a timeline appears in the **Performance** box, above the sliders.



Each of the six rows is one arc. The yellow shape shows when that arc is lit; the red line is the current position, sweeping left to right and looping. The header names the piece being drawn and shows how far through you are — `5:32 / 11:08` above.

Under the timeline, **After this run** chooses what happens at the end of the piece: **↻ Loop the performance** (the default) starts it again straight away; **■ Stop after this run** lets the current pass finish and then stops, with the arcs holding their final state and the music fading out. You can change it while the piece plays — choosing Stop part-way through still lets that pass finish, and the header counts down to the end. The choice lasts until you change it or the box restarts, which puts it back to Loop.

If you pick a different sequence while one is playing, the header says so — "*FAST*" *starts on the next press* (of Play or the sequence button). The drawing keeps showing the piece that is actually playing until you stop and start it again. Nothing has been lost; the box is just refusing to cut a piece off part-way

through.

Before the piece has been started, the header reads *staged: arcs dark until Play or the sequence button*, and the arcs are dark and still (§3). After a Stop it reads *stopped*, and the arcs hold wherever the piece left them.

Note that the sequence controls **when** each arc rises and falls, not how bright or fast it goes. That still comes from the sliders. So you can re-tune the look of the piece at any time without touching the choreography.

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8. The Sequences tab



Four sections.

Active sequence — pick which piece plays. If a sequence is playing when you change this, the new one starts at the **next** press of Play or the sequence button, so you never cut a piece off mid-way. The line under the menu tells you which of the two happened, and the Control tab names the selected piece in its status strip.

Stored sequences — every piece saved on the box, and how many of the 32 slots are in use. Each has two buttons:

- **Edit** loads the piece into *Create / edit* below, without making it the active one.
- **Delete** removes it from the box, after asking you to confirm. This can't be undone, so to keep a copy press **Edit** and then **Export CSV** first. The active sequence shows **active** instead of a Delete button: to delete it, choose a different active sequence first. Deleting a piece that is playing doesn't stop it. It plays on until it is stopped, and is simply gone from the list.

Preview — the same timeline view, for the sequence you selected, so you can see its shape before committing to it. The header gives its number of steps and its loop length.

Create / edit — build a piece of your own:

1. Give it a **name**. Saving with the name of an existing sequence replaces that one, so use a new name if you want to keep both.
2. Add **steps**. A step is a moment in time, a set of arcs and a ramp: "*at 4 minutes, arcs 1, 2 and 3 are on, fading in over 2 seconds.*" Times are in seconds, to a tenth of a second (`240` or `240.5`); the `m:ss.s` beside each time shows the same moment in minutes. Tick the arcs that should be lit at that moment.
3. Set each step's **ramp** — how long the fade into that step takes. The fade *finishes* at the step's time, so a step at 240 s with a 2 s ramp starts changing at 238 s. A ramp of `0` switches instantly. New steps take the **new-step ramp** from the box above.
4. Set the **loop** length — where the piece starts over. Leave it blank and it loops at the last step plus that step's ramp.
5. **Save**. The drawing at the bottom updates as you work, so you can see the shape before saving.

The piece loops. It starts in the state of its last step, so the join is seamless.

Opening from darkness. Because the box waits in the dark (§3), a piece that ends with every arc off, as "Original" does, opens smoothly from black. A piece that ends with arcs lit and has no step at 0 s jumps straight to that last state when Play is pressed. To avoid that, add a first step at **0 s with no arcs ticked**. In a looping piece the arcs then also return to dark at the start of every loop.

To modify an existing piece rather than start blank, press **Edit** beside it under *Stored sequences*. (**Load selected** does the same for the active sequence.)

The line beside the Import / Export buttons shows the step count and the **loop length**. When a music track is loaded (§9) it shows the track's length too, so you can make the two match exactly.

The box holds up to **32 sequences**, each up to **1000 steps**. When every slot is in use, saving a new name fails with a message saying so: delete one you no longer need. Long lists scroll inside the card; more rows appear as you scroll down.

Writing a sequence in a spreadsheet

A long piece — one that follows a 40-minute track, say — is far easier to write in a spreadsheet than one step at a time. Lay it out like this and save it as CSV:

time	arc1	arc2	arc3	arc4	arc5	arc6	ramp
4	1	0	0	0	0	0	4
64	1	1	0	0	0	0	0.5
2:04.5	x	x	x				2
2400	END						

- **time** is seconds (`64`, `64.5`, `2,392.0`) or minutes and seconds (`1:04.5`), to a tenth of a second.
- For each arc, **1**, **x** or **on** means lit; **0**, **off** or an empty cell means dark.
- **ramp** is the fade into that row's state, in seconds (0–600). A blank ramp cell takes the editor's new-step ramp.
- Keep the header row: it's how the importer knows which column is the ramp.
- An optional last row with **END** sets where the loop closes — for a 40-minute track put END at `2400`. Without it, the loop closes at the last step plus its ramp.
- Instead of six arc columns you can use a single column of numbers 0–63 (header `time,mask,ramp`), where arc 1 counts 1, arc 2 counts 2, arc 3 counts 4, and so on.
- A file in the older layout, with no ramp column, still imports: every step then gets the ramp between the last step and END, which is what that layout meant.

Press **Import CSV…** and choose the file. It fills the editor and the drawing, but nothing is sent to the box until you press **Save**. If any row has a problem, nothing is loaded and the lines are listed by number, so you can fix them in the spreadsheet.

Export CSV does the reverse: it downloads what's in the editor in the same layout, ready to open in a spreadsheet and import again.

The built-in sequence, **"Original"**, is the piece the arcs wake one at a time over about five minutes, all six hold together, then they fall away in the same order — an eleven minute loop.

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9. Playing the music with the lights

The **Audio** section of the Control tab's **Performance** box plays a music track from the laptop in step with the sequence, through the laptop's own sound output (and on to the speakers). The track stays on the laptop; it is never sent to the box.

1. On the laptop that feeds the speakers, open the control page and tick **Play audio on this browser**. Phones used only as remotes leave this unticked and stay silent.
2. Press **Choose track…** and pick the music file. The browser keeps a copy, so after a reload or a restart you don't have to pick it again — as long as you open the page at the same address.
3. **Click anywhere on the page** when the yellow banner asks. Browsers won't play sound until someone has clicked on the page, so this is needed once each time the page is opened.

From then on the music follows the lights, however the sequence is started — the Play button on this page, the front-panel button, or another phone. It starts at the same point in the piece as the arcs and fades out when the sequence stops or the box goes to Gallery. Once playing, the music is never adjusted — any jump in it would be heard.

With **When necessary, apply time correction to Slow Arcs during playback** ticked (the default), the page instead nudges the *sequence* a few milliseconds at a time to stay with the music; nobody can see the arcs move 20 ms early or late. Unticked, music and lights simply run side by side from a common start. The status line shows how closely they agree, e.g. `playing (+8 ms) · time correction on`. Tick it on one browser only: two laptops following two copies of the track would pull the sequence between them.

- **Match the lengths.** The sequence's loop length should equal the track's length — a 40:00 track needs a sequence that closes at 2400 s (see the END row in §8). When they differ, a large amber banner across the top of the Performance box says so.
- **If the sound is early or late** compared with the arcs, adjust **offset**: a positive number plays the sound earlier. Bluetooth speakers need several hundred ms; a cable or USB sound interface needs little or none.
- **Use a compressed copy of the track** such as a 256 kbps AAC `.m4a`. It sounds the same as the master and works in every browser. For no loss at all, a FLAC copy is about 40% smaller than the WAV; Chrome and Edge also accept the full-size WAV.
- **Keep the page open and in front**, and set the laptop never to sleep. Turn off notification and alert sounds, or they'll come out of the show speakers too.

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10. Putting the box on the gallery WiFi

Open **WiFi setup** at the bottom of the Control tab.



1. Press **Scan networks** and wait a few seconds.
2. Tap your network in the list. A padlock means it needs a password.
3. Type the password and press **Join**.
4. Watch the status strip at the top. Once the box has joined, it reads *WiFi: connected to* your network with an address, and an **Address** line appears showing `http://slow-arc.local`.

Your phone drops off the `SlowArc-Setup` network about a minute after the join succeeds. Rejoin the gallery WiFi and go to `http://slow-arc.local` — no need to have noted the number.

If the join fails, the `SlowArc-Setup` network stays up — rejoin it and try again. The box will not strand you.

You only ever need to do this once. It is remembered, and the box reconnects by itself after a power cut. **The arcs do not need WiFi to run** — it only matters for changing settings from a phone.

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11. If something looks wrong

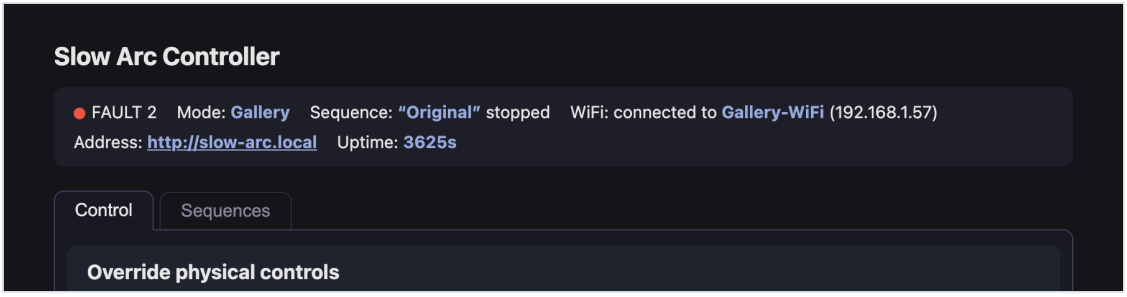
Every arc is dark and still, and the light is blinking blue. Nothing is wrong: the box is in Performance, waiting for the piece to start (§3). Press Play or the sequence button to start it, or switch to Gallery to bring the arcs back to their usual settings.

An arc is dark or not turning. Check its on/off button on the Control tab — it may have been switched off. Then check its sliders aren't at 0. Then check the status light for red flashes.

The page says OFFLINE or won't load. Either the box is powered down or your phone has dropped off the network. Check the box's status light first: if it is lit, the box is fine and the problem is at your end — reconnect and the page recovers by itself. The arcs keep running regardless; the page going away doesn't affect the artwork.

While the page says OFFLINE the sliders are greyed out on purpose. They are showing the last values received, not what the box is doing, so they are not safe to drag.

The status light is flashing red.



Count the flashes (§5) and note the `FAULT` number on the web page. Both say the same thing. Report the number.

Everything is dark after a power cut. Give it a minute. The bulbs fade up slowly and deliberately on every start. If the light then blinks blue, the box was left in Performance and stays dark until the piece is started (see above).

The sequence won't start. Check the mode is Performance — the sequence button does nothing in Gallery mode. **Play** on the page works from either mode.

"Delete failed: that is the active sequence". The box always keeps the sequence the button plays. Choose another one under *Active sequence*, then delete it.

The mode switch seems to do nothing, or the light has a white blip. The page has set the mode since the switch was last moved. The switch takes over again as soon as it is flipped — if it is already where you want it, flip it away and back.

A sequence you saved isn't playing. Saving a sequence doesn't select it. Go to the Sequences tab and choose it under *Active sequence*, then press Play or the sequence button.

Quick reference

Setup network	`SlowArc-Setup` / `slowarc123`
Page address on that network	`http://192.168.4.1`
Page address on gallery WiFi	`http://slow-arc.local` (or the number on the status line)
Healthy light	slow green pulse (Gallery), blue blinking (Performance, waiting, arcs dark), slow blue pulse (sequence playing)
Red 2 / 3 / 4 flashes	bulb / motor / power
White blip every 2 s	web page has overridden the mode switch
Settings saved automatically	yes — brightness, speed, mode, sequences (the music track is kept by the laptop's browser)
Survives a power cut	everything, except arcs switched off from the page

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The screenshots in this guide were taken from the controller's own web page. The readings shown in them — network names, addresses, uptime, slider positions — are examples, not readings from your box.